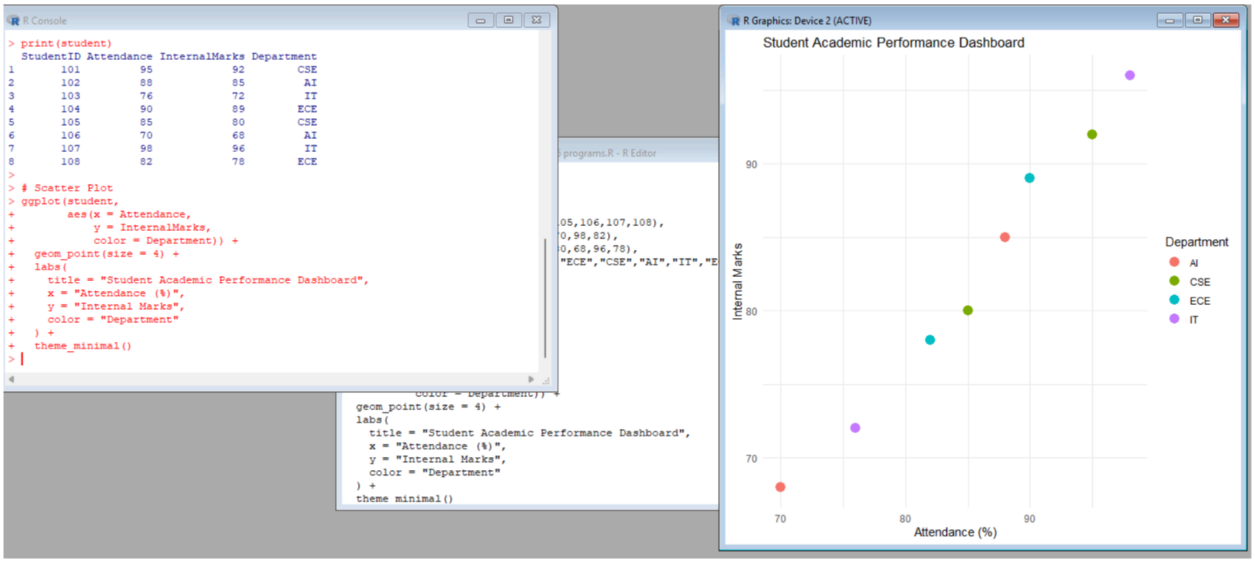
Class Activity: 10.07.2026

Scenario 1: Student Academic Performance Dashboard

R Program code:

```
install.packages("ggplot2")
library(ggplot2)
student <- data.frame(
  StudentID = c(101,102,103,104,105,106,107,108),
  Attendance = c(95,88,76,90,85,70,98,82),
  InternalMarks = c(92,85,72,89,80,68,96,78),
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE")
)
print(student)
ggplot(student,
       aes(x = Attendance,
           y = InternalMarks,
           color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks",
    color = "Department"
  ) +
  theme_minimal()
```

R Program Output:



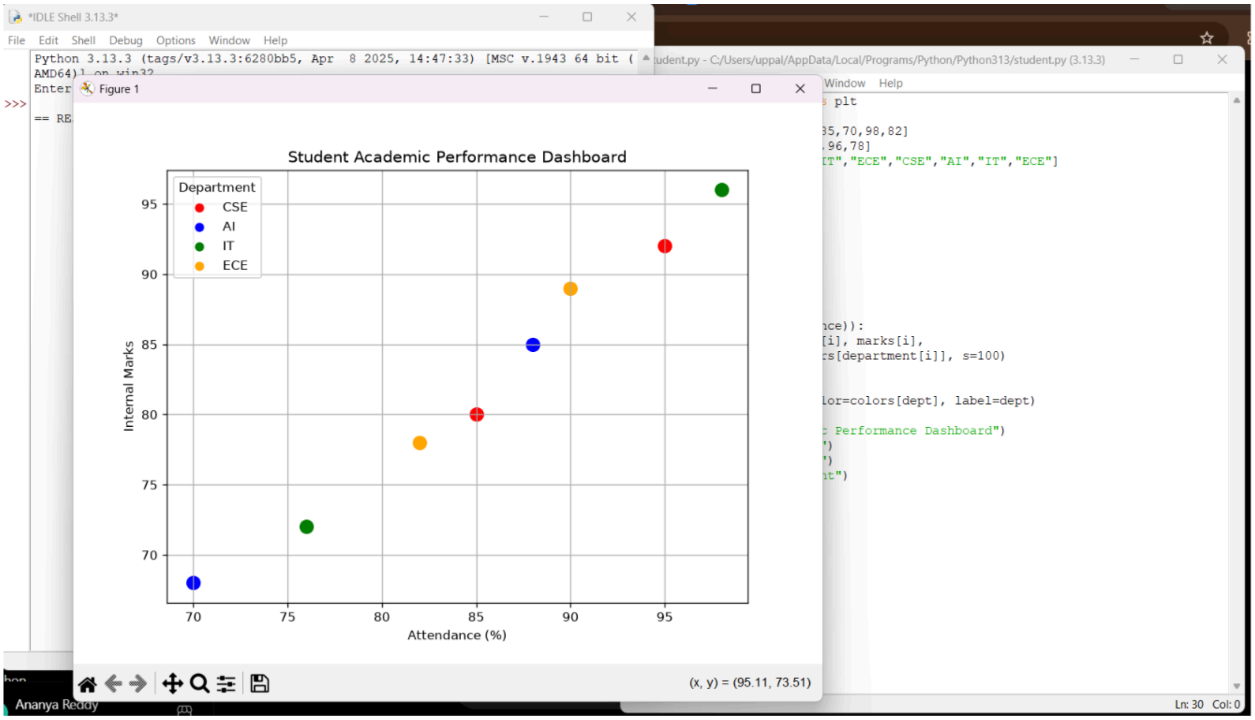
Python code:

```
import matplotlib.pyplot as plt
attendance = [95,88,76,90,85,70,98,82]
marks = [92,85,72,89,80,68,96,78]
department = ["CSE","AI","IT","ECE","CSE","AI","IT","ECE"]
colors = {
    "CSE":"red",
    "AI":"blue",
    "IT":"green",
    "ECE":"orange"
}
plt.figure(figsize=(8,6))
for i in range(len(attendance)):
    plt.scatter(attendance[i],
                marks[i],
                color=colors[department[i]],
                s=100)
for dept in colors:
    plt.scatter([], [], color=colors[dept], label=dept)

plt.title("Student Academic Performance Dashboard")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.legend(title="Department")
plt.grid(True)
```

```
plt.show()
```

Python Program Output:



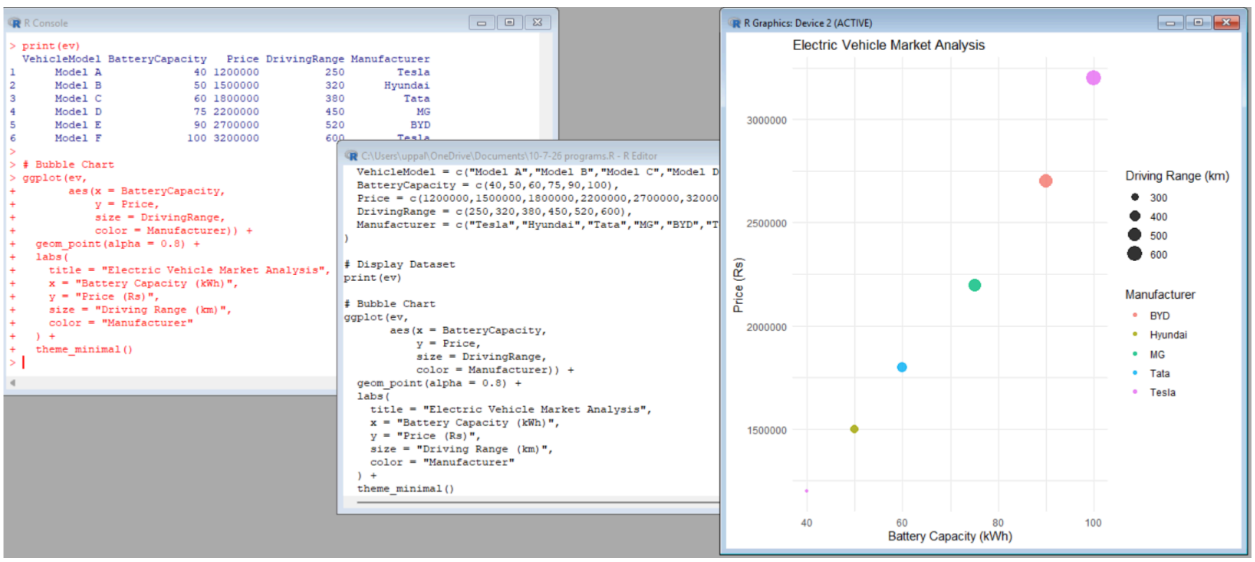
Scenario 2: Electric Vehicle Market Analysis

R Program Code:

```
install.packages("ggplot2")
library(ggplot2)
ev <- data.frame(
  VehicleModel = c("Model A","Model B","Model C","Model D","Model E","Model F"),
  BatteryCapacity = c(40,50,60,75,90,100),
  Price = c(1200000,1500000,1800000,2200000,2700000,3200000),
  DrivingRange = c(250,320,380,450,520,600),
  Manufacturer = c("Tesla","Hyundai","Tata","MG","BYD","Tesla")
)
print(ev)
ggplot(ev,
       aes(x = BatteryCapacity,
           y = Price,
```

```
          size = DrivingRange,
          color = Manufacturer)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Electric Vehicle Market Analysis",
    x = "Battery Capacity (kWh)",
    y = "Price (Rs)",
    size = "Driving Range (km)",
    color = "Manufacturer"
  ) +
  theme_minimal()
```

R Program Output:

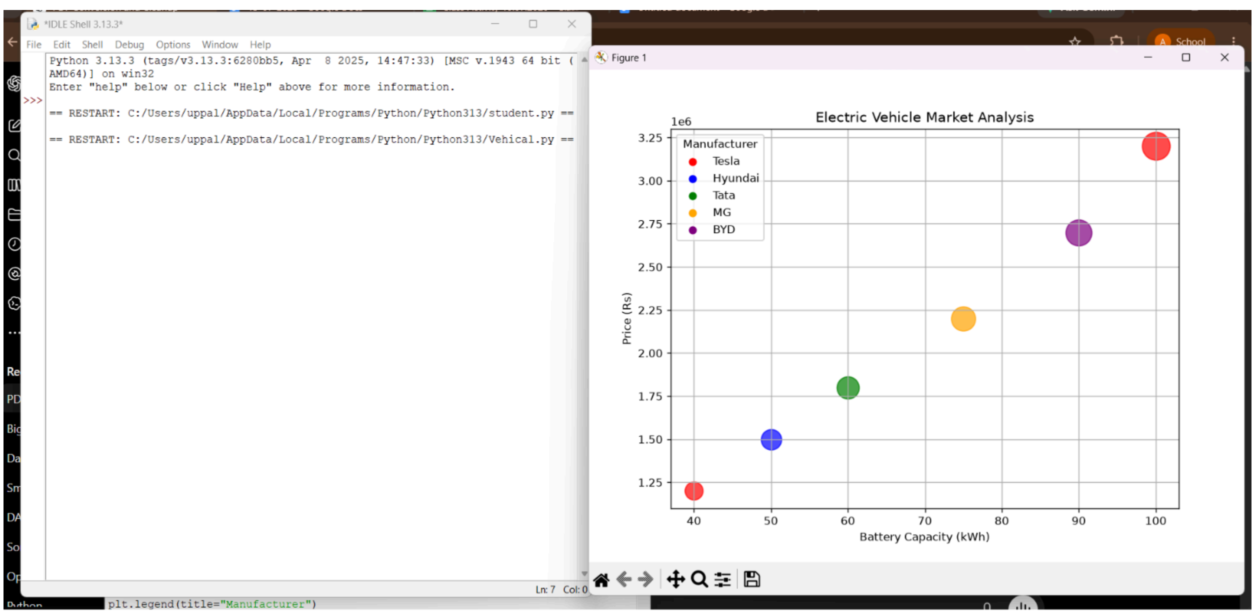


Python Code Program:

```
import matplotlib.pyplot as plt
battery = [40,50,60,75,90,100]
price = [1200000,1500000,1800000,2200000,2700000,3200000]
range_km = [250,320,380,450,520,600]
manufacturer = ["Tesla","Hyundai","Tata","MG","BYD","Tesla"]
colors = {
    "Tesla":"red",
    "Hyundai":"blue",
    "Tata":"green",
    "MG":"orange",
```

```
    "BYD":"purple"
}
plt.figure(figsize=(8,6))
for i in range(len(battery)):
    plt.scatter(
        battery[i],
        price[i],
        s=range_km[i],
        color=colors[manufacturer[i]],
        alpha=0.7
    )
for company in colors:
    plt.scatter([], [], color=colors[company], label=company)
plt.title("Electric Vehicle Market Analysis")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price (Rs)")
plt.legend(title="Manufacturer")
plt.grid(True)
plt.show()
```

Python Program Output:



Scenario 3: Hospital Patient Admission Analysis

R program Code:

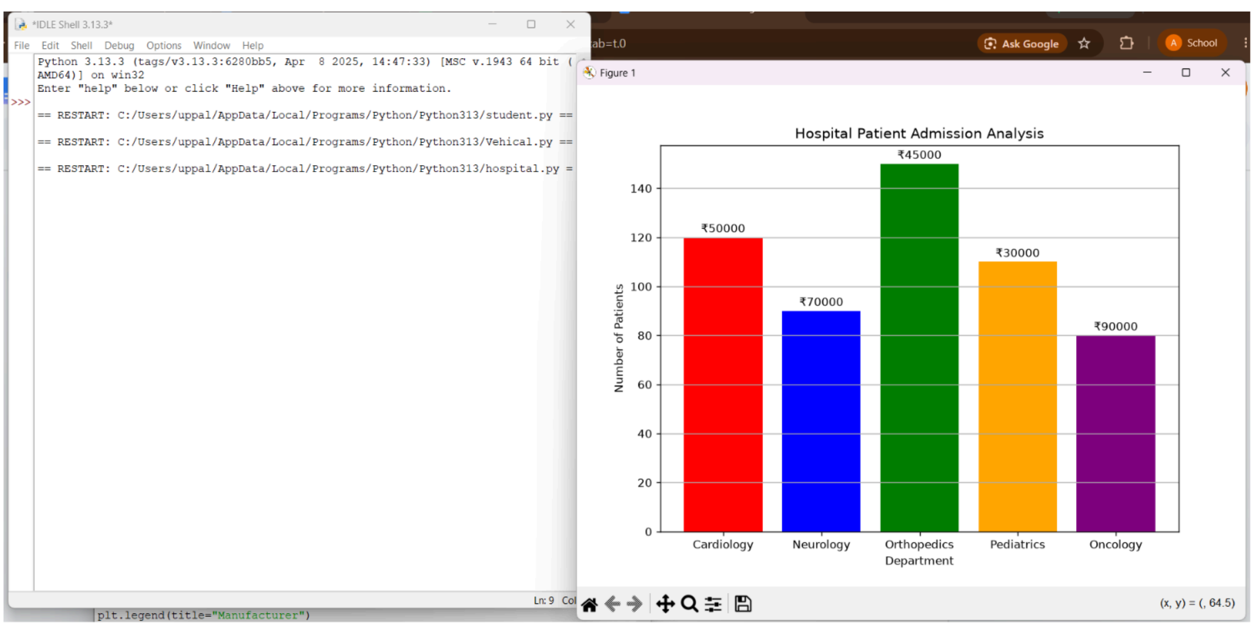
```r
install.packages("ggplot2")
library(ggplot2)
hospital <- data.frame(
  Department = c("Cardiology","Neurology","Orthopedics","Pediatrics","Oncology"),
  NumberOfPatients = c(120,90,150,110,80),
  AverageTreatmentCost = c(50000,70000,45000,30000,90000)
)
print(hospital)
ggplot(hospital,
       aes(x = Department,
           y = NumberOfPatients,
           fill = Department)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = AverageTreatmentCost),
            vjust = -0.5,
            size = 4) +
  labs(
    title = "Hospital Patient Admission Analysis",
    x = "Department",
    y = "Number of Patients",
    fill = "Department"
  ) +
  theme_minimal()
```

R Program Output:

Python Program Code:

```
import matplotlib.pyplot as plt
department = ["Cardiology","Neurology","Orthopedics","Pediatrics","Oncology"]
patients = [120,90,150,110,80]
cost = [50000,70000,45000,30000,90000]
colors = ["red","blue","green","orange","purple"]
plt.figure(figsize=(8,6))
bars = plt.bar(department, patients, color=colors)
for bar, c in zip(bars, cost):
    plt.text(
        bar.get_x() + bar.get_width()/2,
        bar.get_height() + 2,
        f"₹{c}",
        ha='center',
        fontsize=10
    )
plt.title("Hospital Patient Admission Analysis")
plt.xlabel("Department")
plt.ylabel("Number of Patients")
plt.grid(axis='y')
plt.show()
```

Python Program Output:

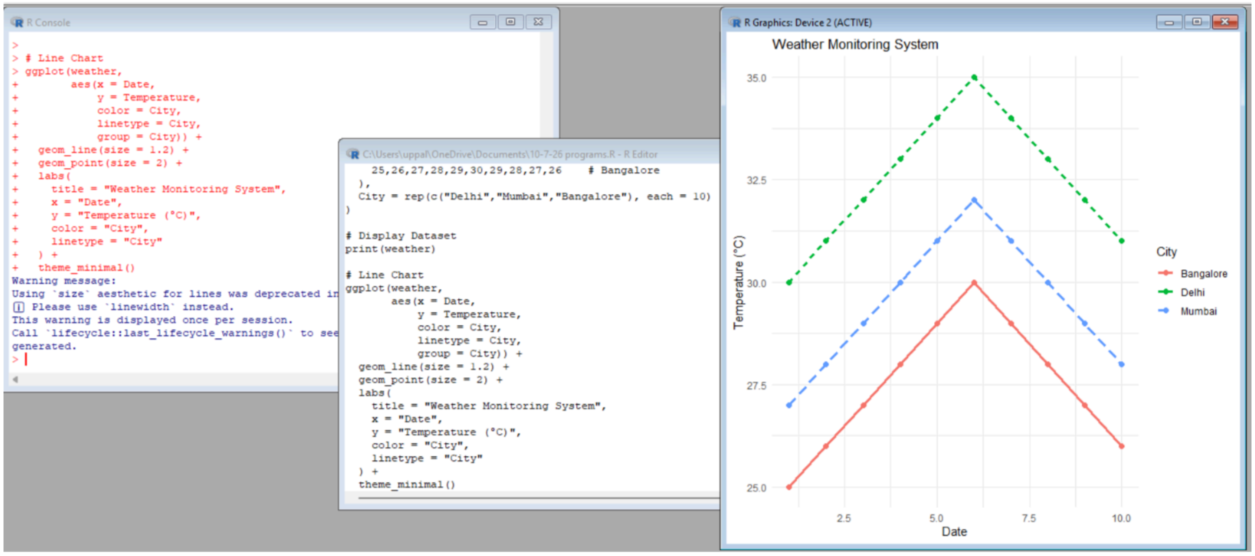


Scenario 4: Weather Monitoring System

R Program Code:

```
import matplotlib.pyplot as plt
dates = list(range(1, 11))
delhi = [30,31,32,33,34,35,34,33,32,31]
mumbai = [27,28,29,30,31,32,31,30,29,28]
bangalore = [25,26,27,28,29,30,29,28,27,26]
plt.figure(figsize=(8,6))
plt.plot(dates, delhi,
        color="red",
        linestyle="-",
        marker="o",
        label="Delhi")
plt.plot(dates, mumbai,
        color="blue",
        linestyle="--",
        marker="s",
        label="Mumbai")
plt.plot(dates, bangalore,
        color="green",
        linestyle=":",
        marker="^",
        label="Bangalore")
plt.title("Weather Monitoring System")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
plt.legend(title="City")
plt.grid(True)
plt.show()
```

R Program Output:

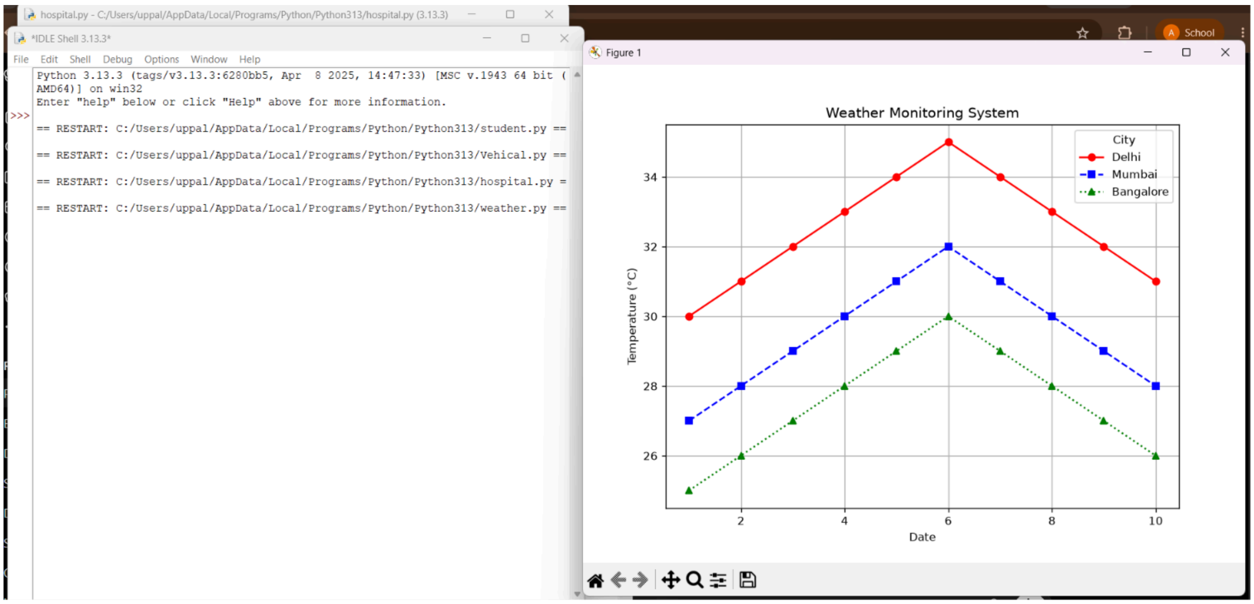


Python Program Code:

```
import matplotlib.pyplot as plt
dates = list(range(1, 11))
delhi = [30,31,32,33,34,35,34,33,32,31]
mumbai = [27,28,29,30,31,32,31,30,29,28]
bangalore = [25,26,27,28,29,30,29,28,27,26]
plt.figure(figsize=(8,6))
plt.plot(dates, delhi,
        color="red",
        linestyle="-",
        marker="o",
        label="Delhi")
plt.plot(dates, mumbai,
        color="blue",
        linestyle="--",
        marker="s",
        label="Mumbai")
plt.plot(dates, bangalore,
        color="green",
        linestyle=":",
        marker="^",
        label="Bangalore")
plt.title("Weather Monitoring System")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
```

```
plt.legend(title="City")
plt.grid(True)
plt.show()
```

Python Program Output:



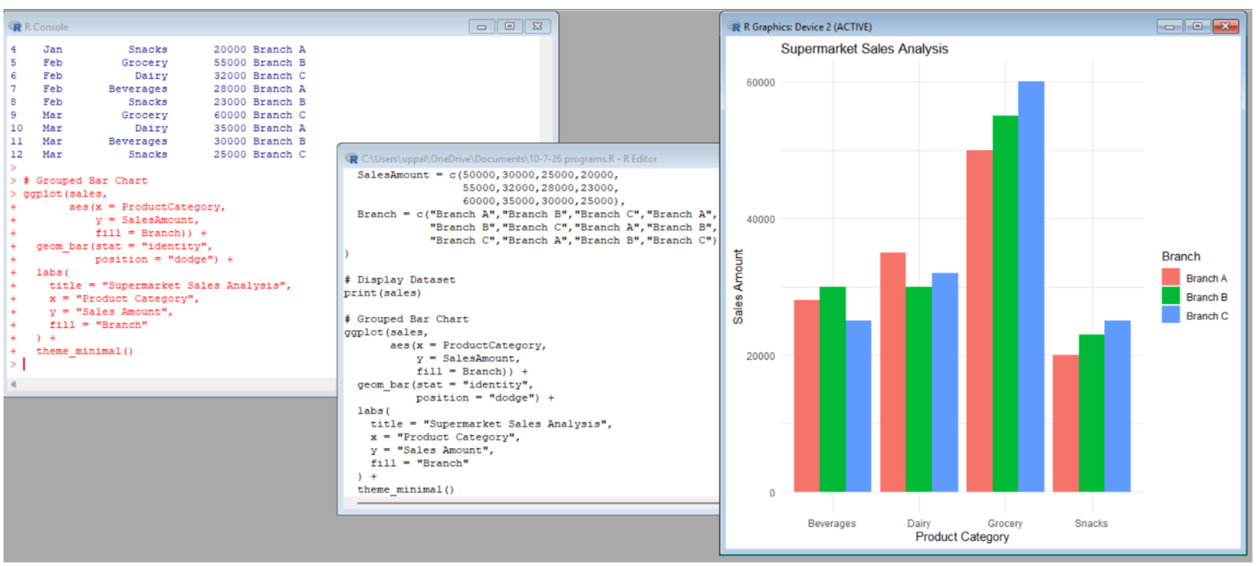
Scenario 5: Supermarket Sales Analysis

R Program Code:

```
install.packages("ggplot2")
library(ggplot2)
sales <- data.frame(
  Month = c("Jan","Jan","Jan","Jan",
           "Feb","Feb","Feb","Feb",
           "Mar","Mar","Mar","Mar"),
  ProductCategory = c("Grocery","Dairy","Beverages","Snacks",
                     "Grocery","Dairy","Beverages","Snacks",
                     "Grocery","Dairy","Beverages","Snacks"),
  SalesAmount = c(50000,30000,25000,20000,
                 55000,32000,28000,23000,
                 60000,35000,30000,25000),
  Branch = c("Branch A","Branch B","Branch C","Branch A",
            "Branch B","Branch C","Branch A","Branch B",
```

```
            "Branch C","Branch A","Branch B","Branch C")
)
print(sales)
ggplot(sales,
       aes(x = ProductCategory,
           y = SalesAmount,
           fill = Branch)) +
  geom_bar(stat = "identity",
           position = "dodge") +
  labs(
    title = "Supermarket Sales Analysis",
    x = "Product Category",
    y = "Sales Amount",
    fill = "Branch"
  ) +
  theme_minimal()
```

R Program Output:

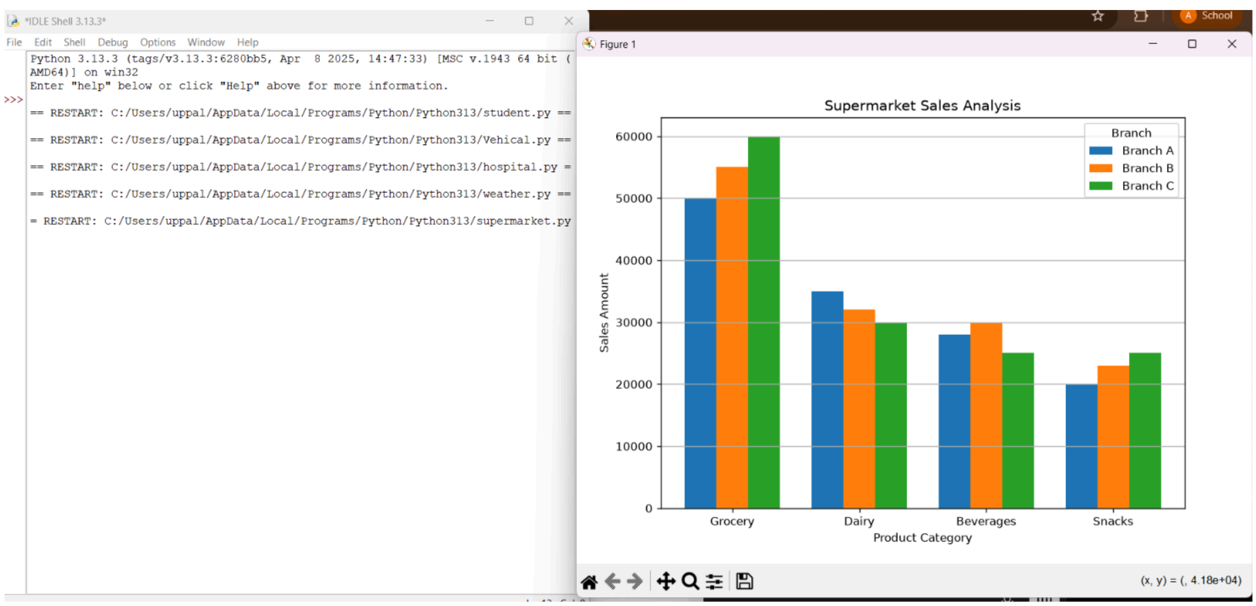


Python Program Code:

```
import matplotlib.pyplot as plt
import numpy as np
categories = ["Grocery", "Dairy", "Beverages", "Snacks"]
branchA = [50000, 35000, 28000, 20000]
branchB = [55000, 32000, 30000, 23000]
```

```
branchC = [60000, 30000, 25000, 25000]
x = np.arange(len(categories))
width = 0.25
plt.figure(figsize=(8,6))
plt.bar(x - width, branchA, width, label="Branch A")
plt.bar(x, branchB, width, label="Branch B")
plt.bar(x + width, branchC, width, label="Branch C")
plt.title("Supermarket Sales Analysis")
plt.xlabel("Product Category")
plt.ylabel("Sales Amount")
plt.xticks(x, categories)
plt.legend(title="Branch")
plt.grid(axis="y")
plt.show()
```

Python Program Output:



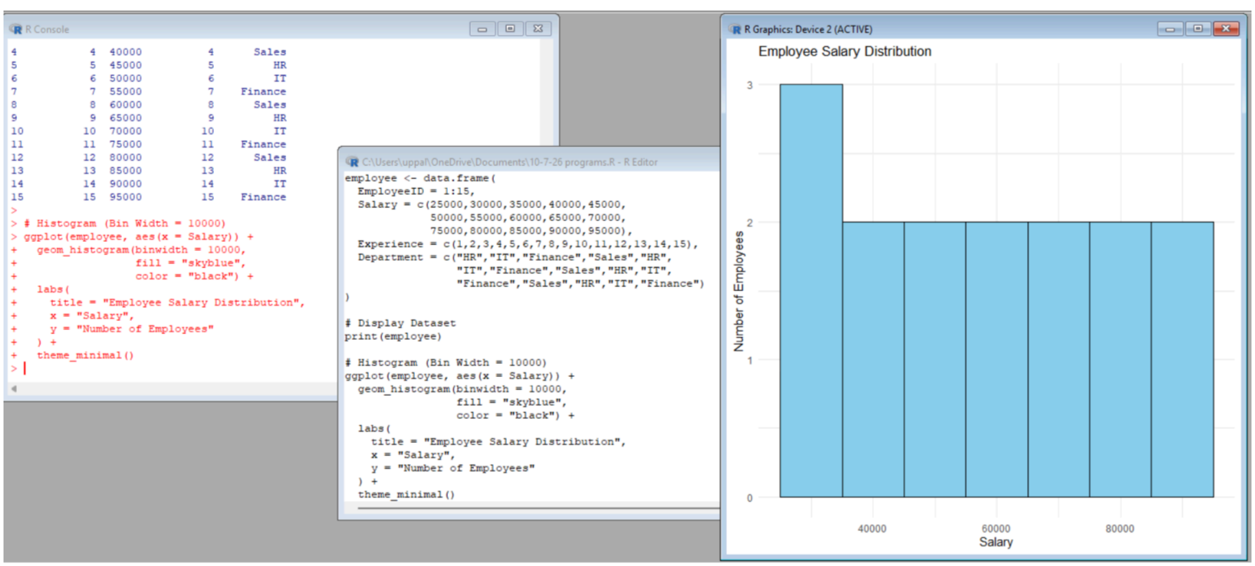
Scenario 6: Employee Salary Distribution

R Program Code:

```
install.packages("ggplot2")
library(ggplot2)
employee <- data.frame(
  EmployeeID = 1:15,
```

```
  Salary = c(25000,30000,35000,40000,45000,
             50000,55000,60000,65000,70000,
             75000,80000,85000,90000,95000),
  Experience = c(1,2,3,4,5,6,7,8,9,10,11,12,13,14,15),
  Department = c("HR","IT","Finance","Sales","HR",
                 "IT","Finance","Sales","HR","IT",
                 "Finance","Sales","HR","IT","Finance")
)
print(employee)
ggplot(employee, aes(x = Salary)) +
  geom_histogram(binwidth = 10000,
                 fill = "skyblue",
                 color = "black") +
  labs(
    title = "Employee Salary Distribution",
    x = "Salary",
    y = "Number of Employees"
  ) +
  theme_minimal()
```

R Program Output:

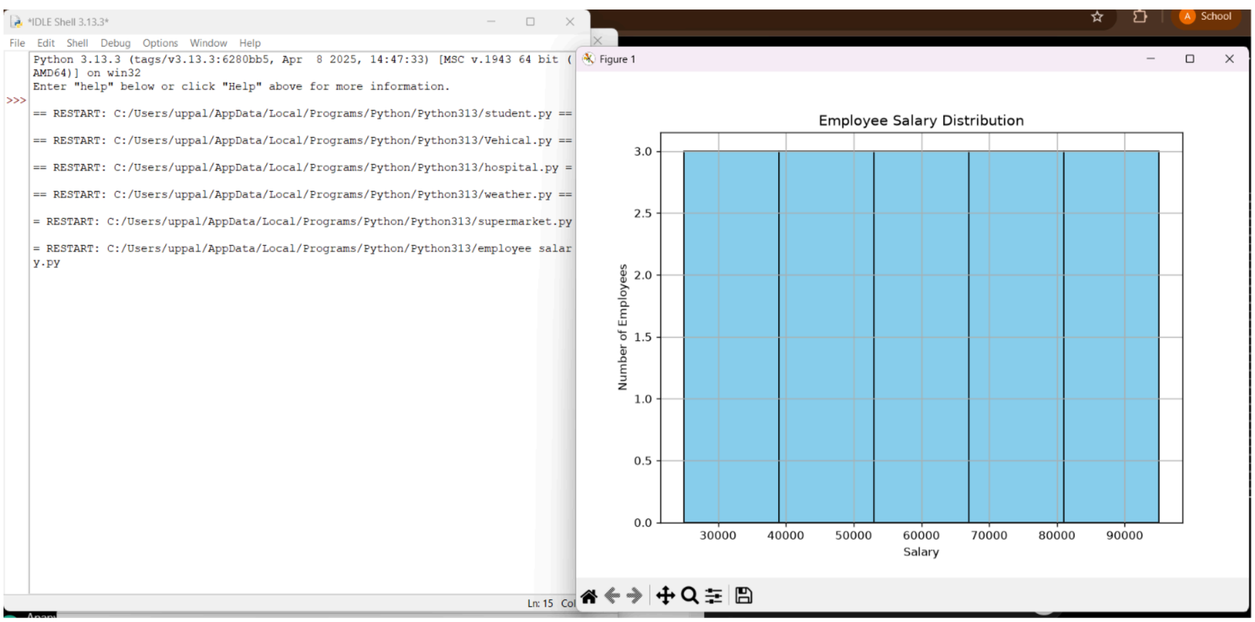


Python Program Code:

```
import matplotlib.pyplot as plt
salary = [25000,30000,35000,40000,45000,
```

```
        50000,55000,60000,65000,70000,
        75000,80000,85000,90000,95000]
plt.figure(figsize=(8,6))
plt.hist(
    salary,
    bins=5,
    color="skyblue",
    edgecolor="black"
)
plt.title("Employee Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Number of Employees")
plt.grid(True)
plt.show()
```

Python Program Output:



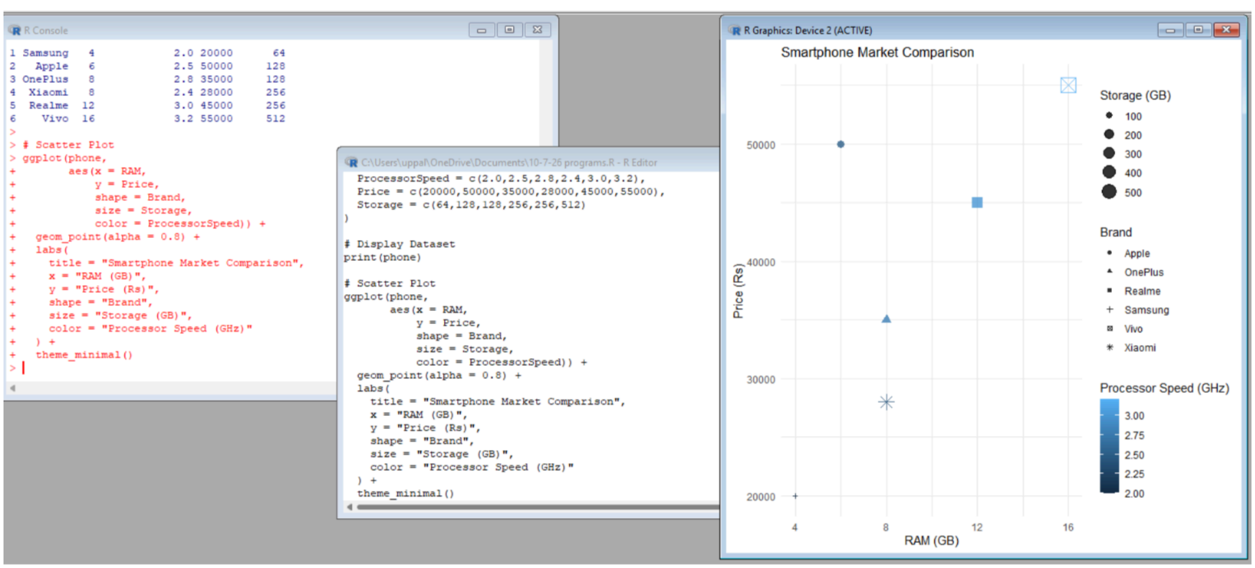
Scenario 7: Smartphone Market Comparison

R Program Code:

```
install.packages("ggplot2")
library(ggplot2)
phone <- data.frame(
  Brand = c("Samsung","Apple","OnePlus","Xiaomi","Realme","Vivo"),
```

```
  RAM = c(4,6,8,8,12,16),
  ProcessorSpeed = c(2.0,2.5,2.8,2.4,3.0,3.2),
  Price = c(20000,50000,35000,28000,45000,55000),
  Storage = c(64,128,128,256,256,512)
)
print(phone)
ggplot(phone,
       aes(x = RAM,
           y = Price,
           shape = Brand,
           size = Storage,
           color = ProcessorSpeed)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Smartphone Market Comparison",
    x = "RAM (GB)",
    y = "Price (Rs)",
    shape = "Brand",
    size = "Storage (GB)",
    color = "Processor Speed (GHz)"
  ) +
  theme_minimal()
```

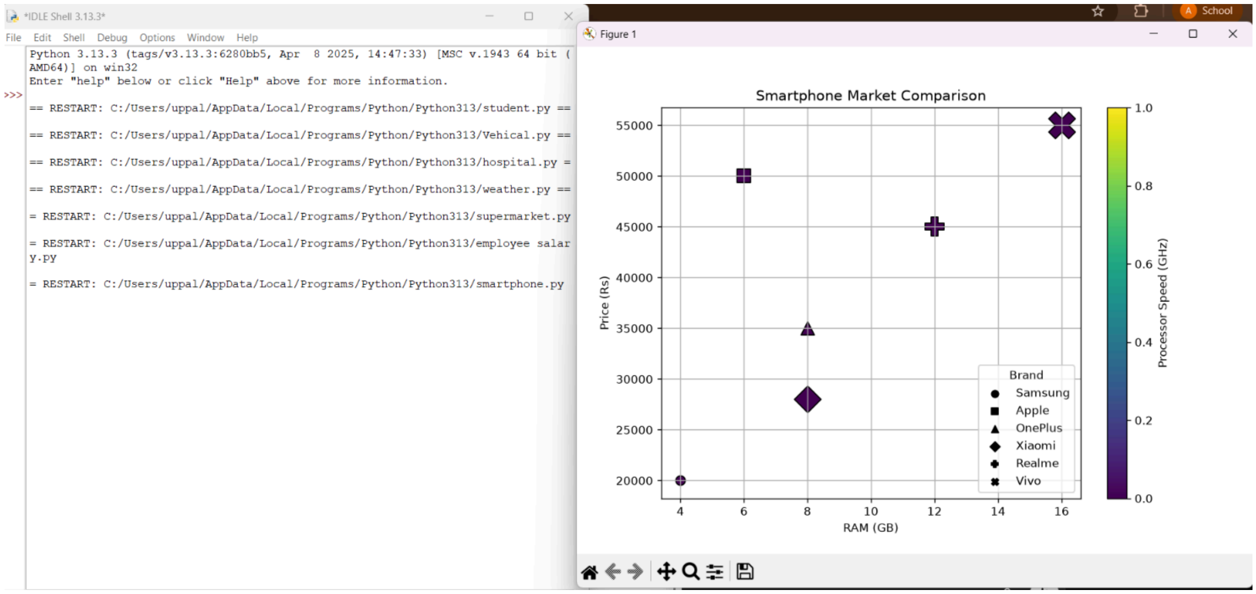
R program Output:



Python Program Code:

```
import matplotlib.pyplot as plt
brand = ["Samsung","Apple","OnePlus","Xiaomi","Realme","Vivo"]
ram = [4,6,8,8,12,16]
processor = [2.0,2.5,2.8,2.4,3.0,3.2]
price = [20000,50000,35000,28000,45000,55000]
storage = [64,128,128,256,256,512]
markers = {
    "Samsung":"o",
    "Apple":"s",
    "OnePlus":"^",
    "Xiaomi":"D",
    "Realme":"P",
    "Vivo":"X"
}
plt.figure(figsize=(8,6))
for i in range(len(brand)):
    plt.scatter(
        ram[i],
        price[i],
        s=storage[i],
        c=[processor[i]],
        cmap="viridis",
        marker=markers[brand[i]],
        edgecolors="black"
    )
for b in markers:
    plt.scatter([], [], marker=markers[b], color="black", label=b)
plt.colorbar(label="Processor Speed (GHz)")
plt.legend(title="Brand")
plt.title("Smartphone Market Comparison")
plt.xlabel("RAM (GB)")
plt.ylabel("Price (Rs)")
plt.grid(True)
plt.show()
```

Python Program Output:



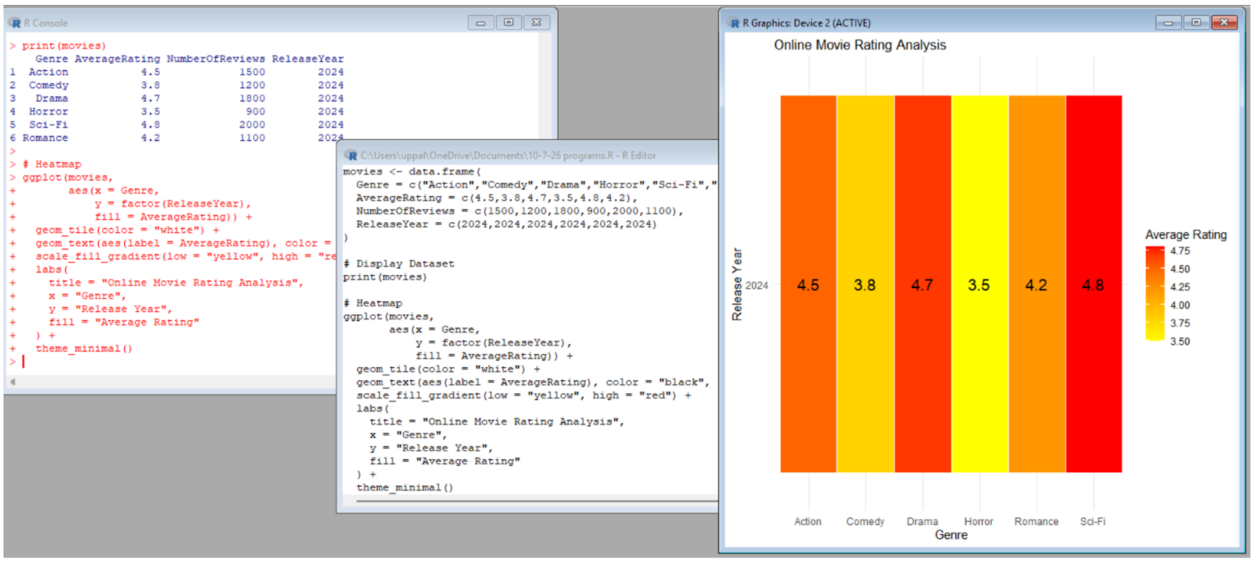
Scenario 8: Online Movie Rating Analysis

R Program Code:

```
install.packages("ggplot2")
library(ggplot2)
movies <- data.frame(
  Genre = c("Action","Comedy","Drama","Horror","Sci-Fi","Romance"),
  AverageRating = c(4.5,3.8,4.7,3.5,4.8,4.2),
  NumberOfReviews = c(1500,1200,1800,900,2000,1100),
  ReleaseYear = c(2024,2024,2024,2024,2024,2024)
)
print(movies)
ggplot(movies,
       aes(x = Genre,
           y = factor(ReleaseYear),
           fill = AverageRating)) +
  geom_tile(color = "white") +
  geom_text(aes(label = AverageRating), color = "black", size = 5) +
  scale_fill_gradient(low = "yellow", high = "red") +
  labs(
    title = "Online Movie Rating Analysis",
    x = "Genre",
    y = "Release Year",
    fill = "Average Rating"
```

```
) +
theme_minimal()
```

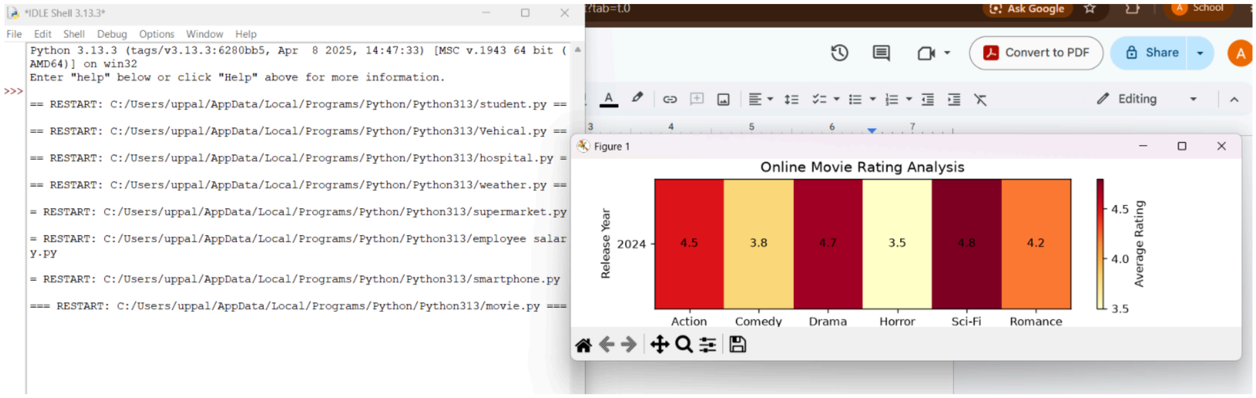
R program Output:



Python Program Code:

```
import matplotlib.pyplot as plt
import numpy as np
genres = ["Action","Comedy","Drama","Horror","Sci-Fi","Romance"]
ratings = [4.5,3.8,4.7,3.5,4.8,4.2]
data = np.array([ratings])
plt.figure(figsize=(8,2))
plt.imshow(data, cmap="YlOrRd", aspect="auto")
plt.xticks(range(len(genres)), genres)
plt.yticks([0], ["2024"])
for i in range(len(genres)):
    plt.text(i, 0, ratings[i],
             ha='center',
             va='center',
             color='black')
plt.colorbar(label="Average Rating")
plt.title("Online Movie Rating Analysis")
plt.xlabel("Genre")
plt.ylabel("Release Year")
plt.show()
```

Python Program Output:



Scenario 9: COVID-19 State-wise Analysis

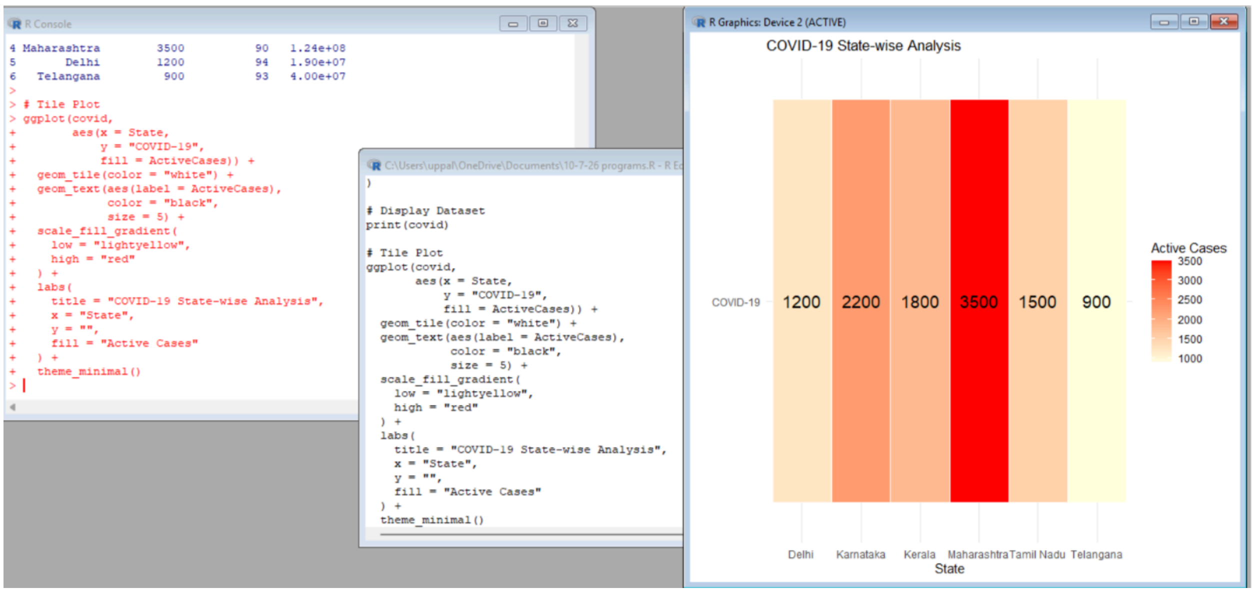
R Program Code:

```
install.packages("ggplot2")

library(ggplot2)
covid <- data.frame(
  State = c("Tamil Nadu","Karnataka","Kerala","Maharashtra","Delhi","Telangana"),
  ActiveCases = c(1500,2200,1800,3500,1200,900),
  Vaccination = c(95,92,96,90,94,93),
  Population = c(72000000,68000000,35000000,124000000,19000000,40000000)
)
print(covid)
ggplot(covid,
       aes(x = State,
           y = "COVID-19",
           fill = ActiveCases)) +
  geom_tile(color = "white") +
  geom_text(aes(label = ActiveCases),
            color = "black",
            size = 5) +
  scale_fill_gradient(
    low = "lightyellow",
    high = "red"
  ) +
```

```
labs(
  title = "COVID-19 State-wise Analysis",
  x = "State",
  y = "",
  fill = "Active Cases"
) +
theme_minimal()
```

R Program Output:

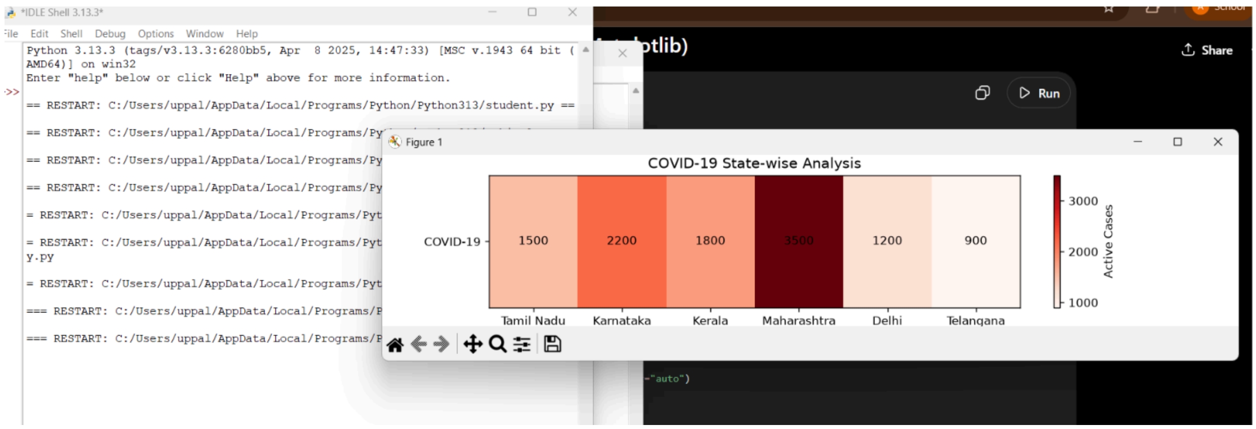


Python Program Code:

```
import matplotlib.pyplot as plt
import numpy as np
states = ["Tamil Nadu","Karnataka","Kerala",
          "Maharashtra","Delhi","Telangana"]
active_cases = [1500,2200,1800,3500,1200,900]
data = np.array([active_cases])
plt.figure(figsize=(10,2))
plt.imshow(data, cmap="Reds", aspect="auto")
plt.xticks(range(len(states)), states)
plt.yticks([0], ["COVID-19"])
for i in range(len(states)):
    plt.text(i, 0,
             active_cases[i],
```

```
            ha="center",
            va="center",
            color="black")
plt.colorbar(label="Active Cases")
plt.title("COVID-19 State-wise Analysis")
plt.xlabel("State")
plt.ylabel("")
plt.show()
```

Python Program Output:



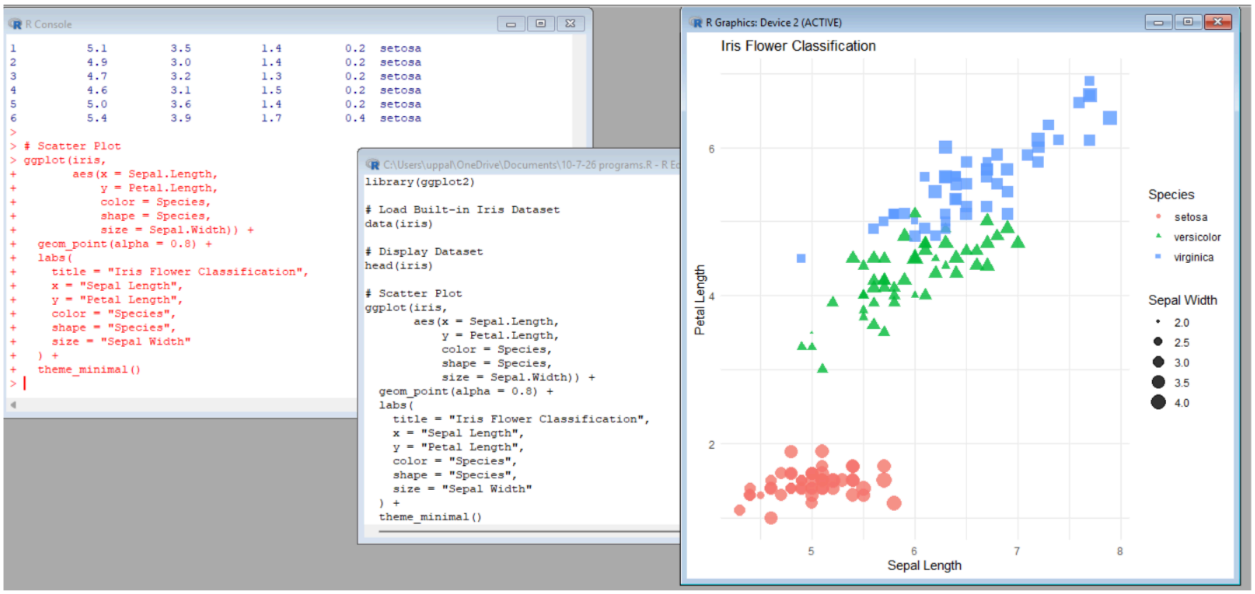
Scenario 10: Iris Flower Classification

R Program Code:

```
install.packages("ggplot2")
library(ggplot2)
data(iris)
head(iris)
ggplot(iris,
       aes(x = Sepal.Length,
           y = Petal.Length,
           color = Species,
           shape = Species,
           size = Sepal.Width)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Iris Flower Classification",
    x = "Sepal Length",
```

```
    y = "Petal Length",
    color = "Species",
    shape = "Species",
    size = "Sepal Width"
  ) +
  theme_minimal()
```

R Program Output:



Python Program Code:

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
iris = load_iris()
sepal_length = iris.data[:, 0]
sepal_width = iris.data[:, 1]
petal_length = iris.data[:, 2]
species = iris.target
species_names = iris.target_names
colors = ["red", "green", "blue"]
markers = ["o", "s", "^"]
plt.figure(figsize=(8,6))
for i in range(3):
    plt.scatter(
        sepal_length[species == i],
```

```
        petal_length[species == i],
        s=sepal_width[species == i] * 40,
        c=colors[i],
        marker=markers[i],
        label=species_names[i],
        alpha=0.7
    )
plt.title("Iris Flower Classification")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.legend(title="Species")
plt.grid(True)
plt.show()
```

Python Program Output: